Diagnostic check: six questions, 10 minutes

Answer without looking at your notes. Indicate the questions for which help is needed. This diagnostic test is not graded.

1. Derivative of x^2 at 3?
2. Domain of $\dfrac{1}{x-2}$?
3. Sign of $(x-1)(x+2)$?
4. Expand $(x^2+1)(x-2)$.
5. For $f(x)=x^2$, compare $f(-x)$ and $f(x)$.
6. If $f(1)=2$, $f'(1)=0$, equation of the tangent?

After the diagnostic check. Go over the flagged concepts with the teacher; redo a similar question, then explain the rule used.

Oral work and logic. Present a variation table, distinguishing the sign of f from the sign of its derivative. Explain why the endpoints of an optimisation problem must be examined.

Investigate before formalising

Can an increasing function have a horizontal tangent? Look for an example; can it have a strictly negative derivative at an interior point?

Organisation. Individual work 5 min, discussion in pairs 5 min, whole-class sharing 5 min. Keep a record of an attempt, even an unfinished one. A conjecture must then be justified.